

T16 Remote Controller Training





T16 Remote Controller Overview

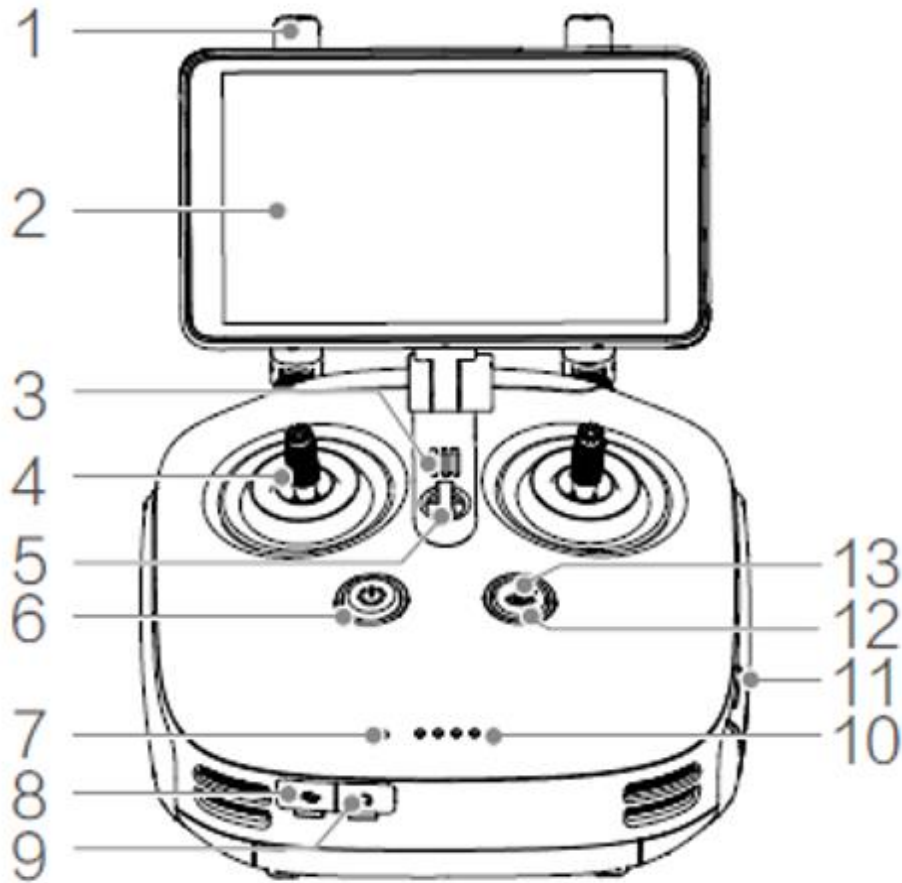
T16 Remote Controller Link Diagram

Disassembling the T16 Remote Controller

Assembling the T16 Remote Controller T16

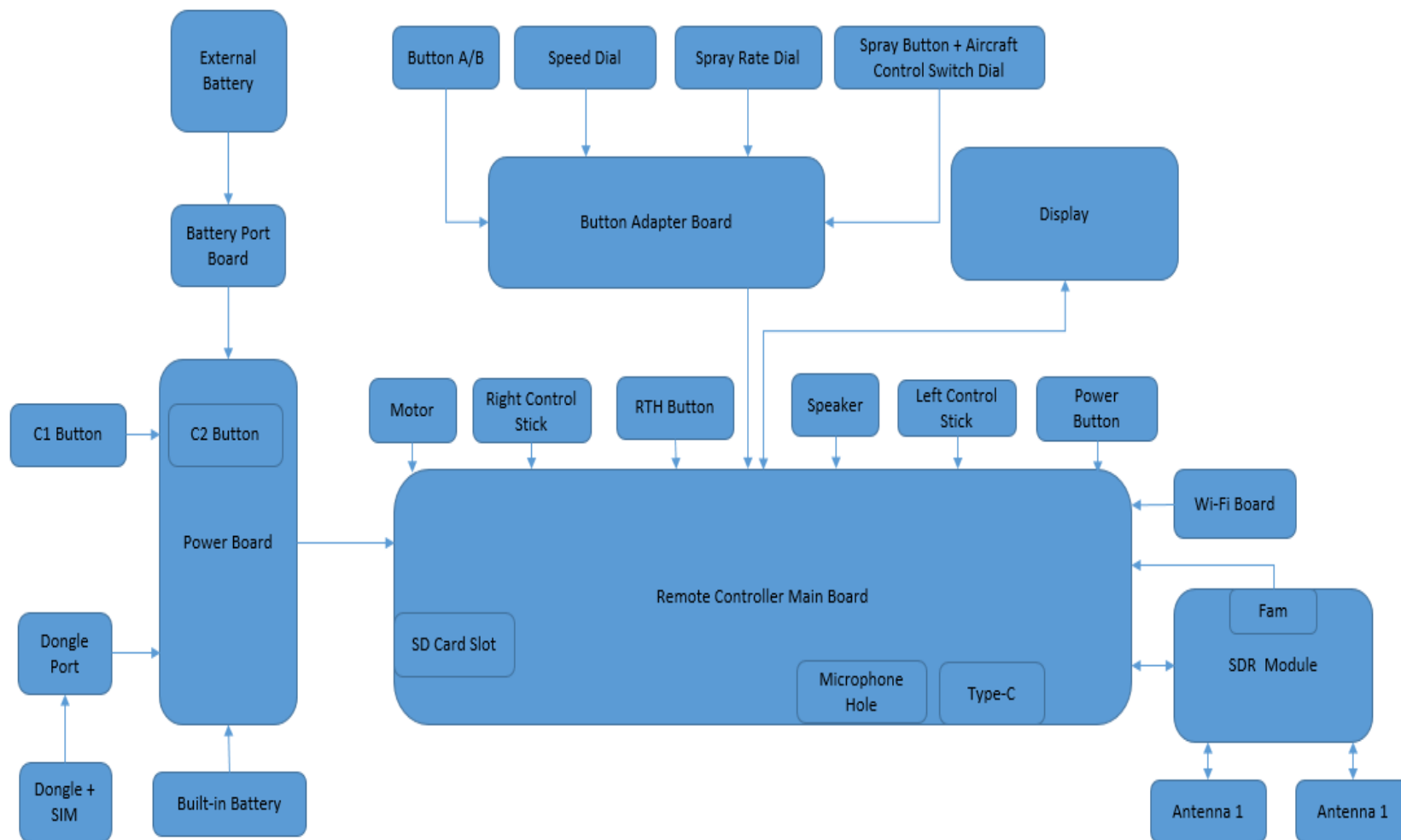
T16 Remote Controller Troubleshooting

T16 Remote Controller Overview



1. Antennas (Relays aircraft control and image transmission signals.)
2. Display Device (Android-based device to run DJI MG2.)
3. Speaker (Audio output)
4. Control Sticks (Controls aircraft movement. Can be set to Mode 1, Mode 2, or Mode 3.)
5. Lanyard Attachment (Used to attach the remote controller lanyard.)
6. Power Button (Used to turn the remote controller on and off.)
7. Status LED (Indicates whether the remote controller is linked to the aircraft.)
8. USB-C Port (Connects to a computer via a USB-C cable for configuration. Connects to the aircraft via a USB-C OTG cable and a Micro USB cable for aircraft firmware update.)
9. 3.5 mm Audio Jack (Used to connect an audio input or output device.)
10. Battery Level LEDs (Displays current battery level.)
11. microSD Card Slot (Provides display device with up to 128 GB of extra storage.)
12. RTH Indicator (Indicates the aircraft RTH status)
13. RTH Button (Press and hold this button to initiate RTH.)

T16 Remote Controller Link Diagram



Remote Controller Main Component Operation Principles

Remote Controller Main Board

Processes all input and output signals of the remote controller.
Controls the remote controller power supply and charging.

Adapter Board

Transfer the signals of buttons and dials.

Wi-Fi Board

Controls the remote controller wireless network function.

SDR Board

Controls the remote controller image transmission function.

Power Board

Power processing module.

Disassembling the T16 Remote Controller



1. Remove the two antennas' metal rings and detach the antennas.



2. Open the dongle cover and remove the left and right bottom rubber pads.

Disassembling the T16 Remote Controller



1. Remove the eight screws on the remote controller rear shell with the Phillips screwdriver.
2. There are two screws on each side (left, right, up and down)

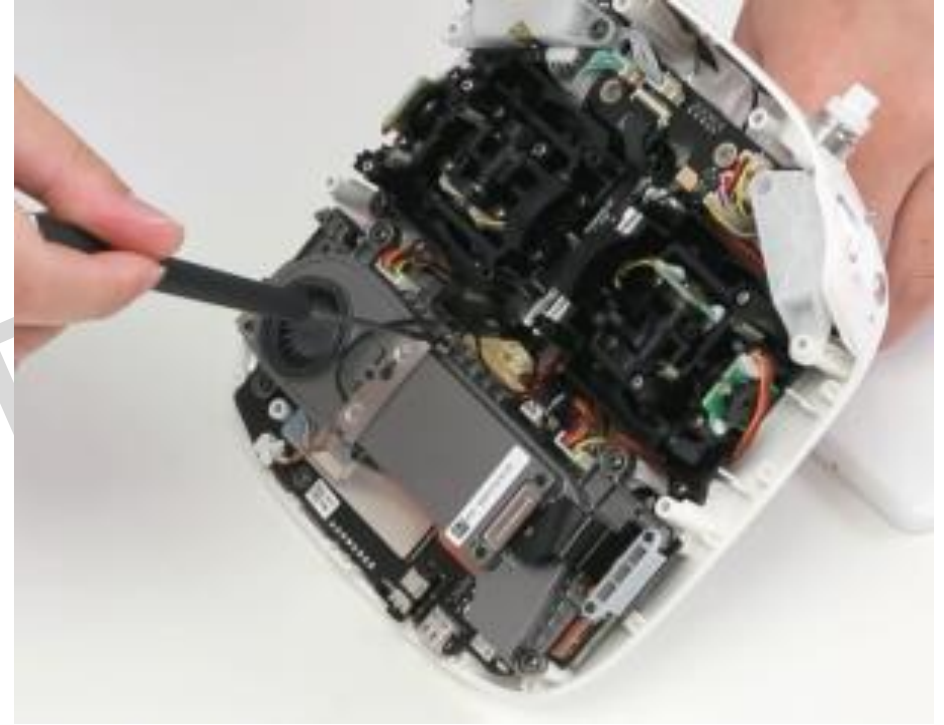


1. Separate the front and rear shell: There are buckles on the top and bottom of the front and rear shells. When separating the shells, press the rear shell and pull it back slightly. Otherwise, it is easy to damage the FPC cable between the front and rear shells.

Disassembling the T16 Remote Controller



1. Use the Phillips screwdriver to remove the two screws securing the fixing piece of the FPC flat cable which connects the power board and main board.
2. Remove the fixing piece and detach the FPC.



1. Use the Phillips screwdriver to remove the five screws (three screws on the left mounting piece and two on the right piece) securing the left and right mounting pieces.

Disassembling the T16 Remote Controller

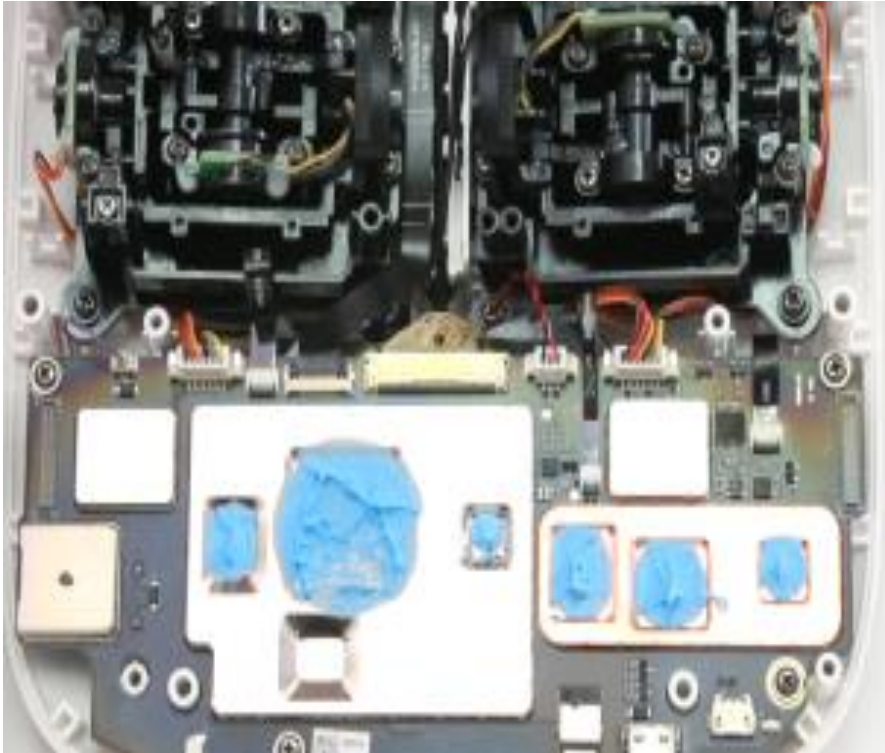


1. Remove the rubber stoppers on the antenna ports with the tweezers and disconnect the SDR antenna and FPC.

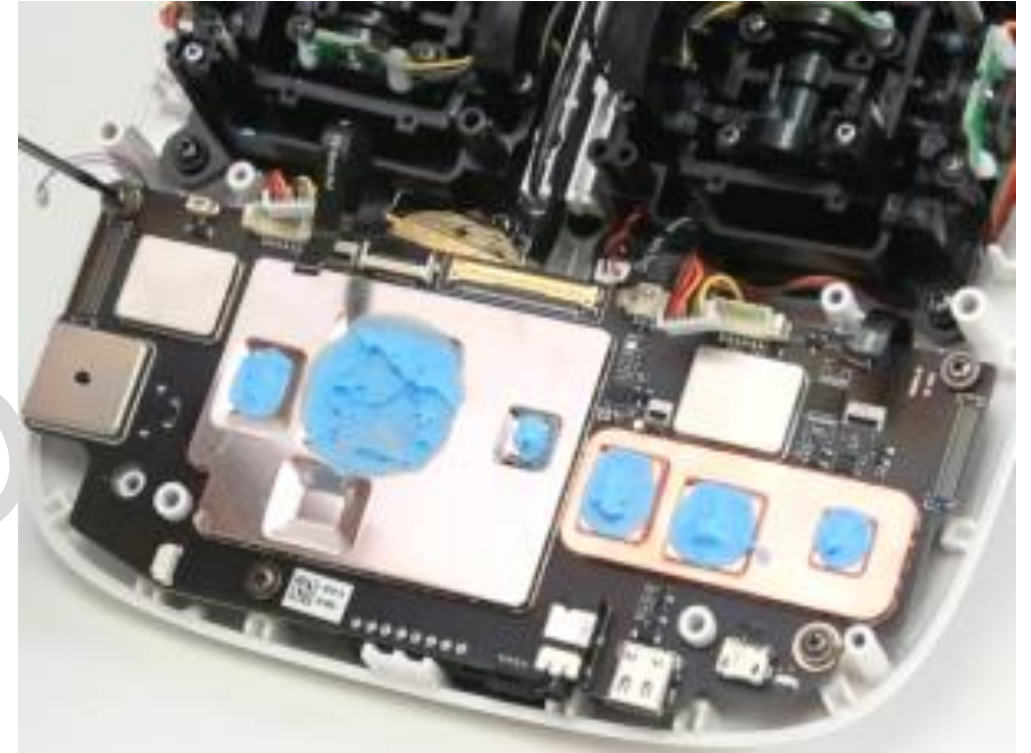


1. Use the Phillips screwdriver to remove the four screws securing the heat sink. Disconnect the fan power cable and remove the heat sink and SDR module.
Do not damage the FPC on the right side.

Disassembling the T16 Remote Controller



1. Remove all FPCs and flat cables on the main board with the tweezers.
2. Use the tweezers to disconnect the motor cable, right control stick cable, RTH button flat cable, button board flat cable, screen flat cable, microphone cable, power cable, left control stick cable, and Wi-Fi cable.

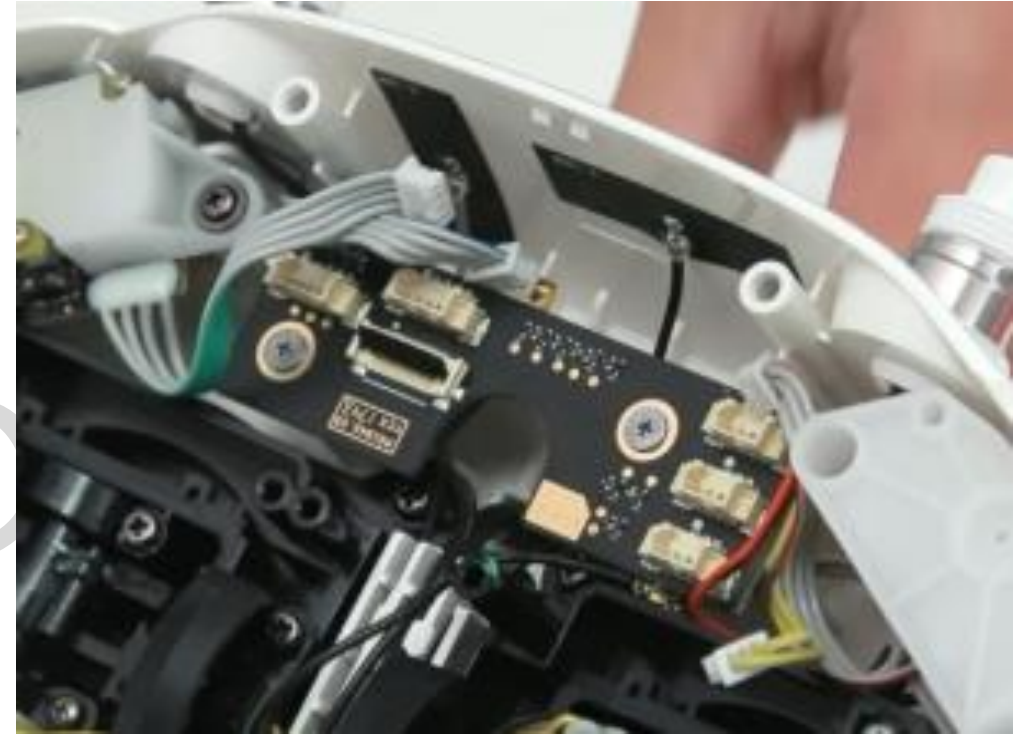


1. Use the Phillips screwdriver to remove the four screws securing the remote controller main board.

Disassembling the T16 Remote Controller



1. Use the Phillips screwdriver to remove the two screws securing the Wi-Fi antenna mounting piece, then remove the antenna.
2. Use the Phillips screwdriver to remove the two screws securing the Wi-Fi board, and then remove the Wi-Fi board.

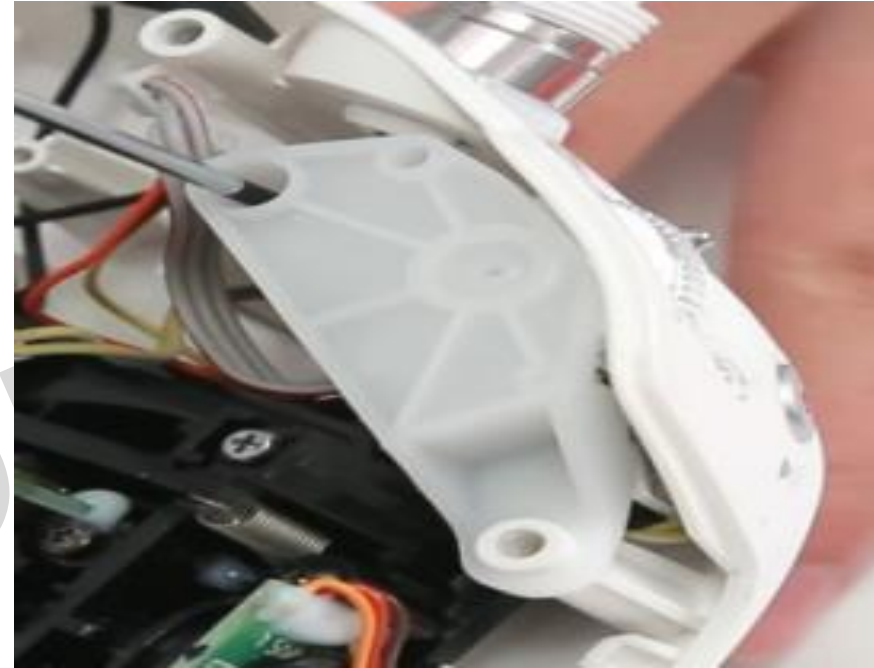


1. Use the Phillips screwdriver to remove the screw securing the speed dial, and then remove the speed dial module.

Disassembling the T16 Remote Controller

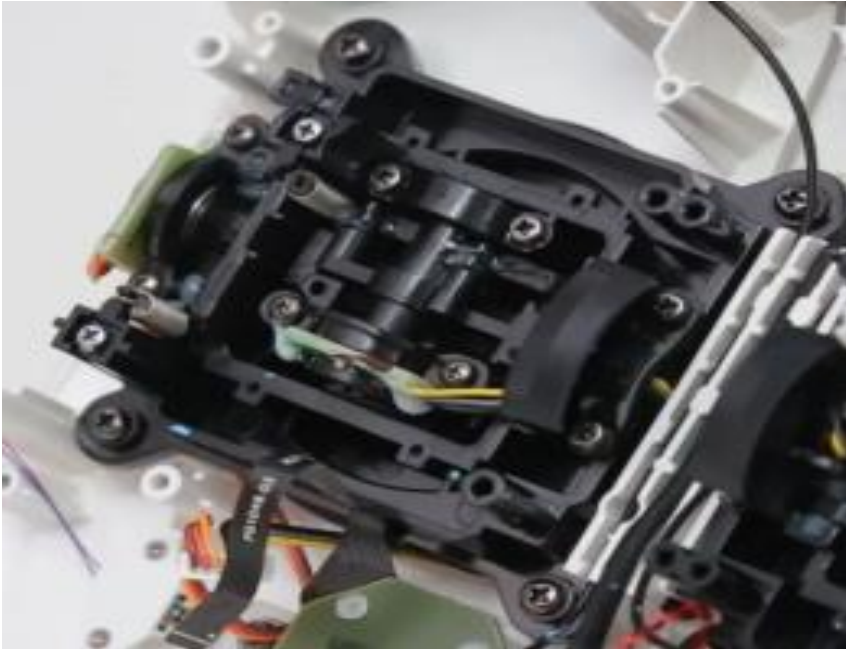


1. Use the Phillips screwdriver to remove the two screws securing the A/B button module, then remove the A/B button module.



1. Use the Phillips screwdriver to remove the two screws securing the A/B button module, then remove the A/B button module.

Disassembling the T16 Remote Controller



1. Use the Phillips screwdriver to remove the eight screws securing the two control stick modules, then remove the two modules.



1. Remove the attitude lever module.

Disassembling the T16 Remote Controller



1. Remove the antenna and screen flat cable from the cable slot and use the Phillips screwdriver to remove the two screws securing the cable slot, then remove the cable slot.

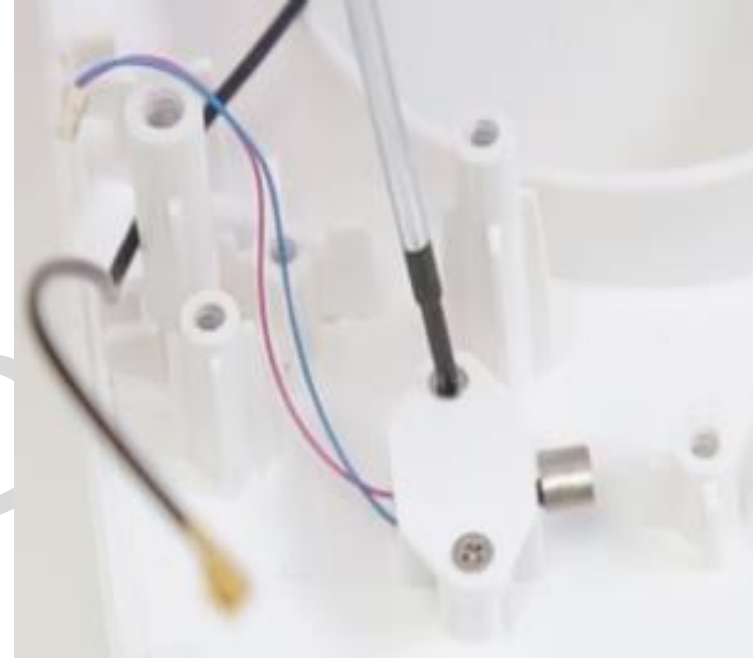


1. Use the Phillips screwdriver to remove the two screws securing the screen module, then remove the screen module. Do not damage the flat cable on the screen module.

Disassembling the T16 Remote Controller

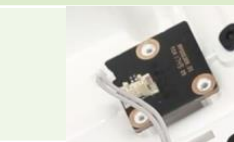


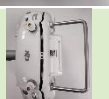
1. Use the Phillips screwdriver to remove the two screws securing the RTH button, then remove the RTH button module.



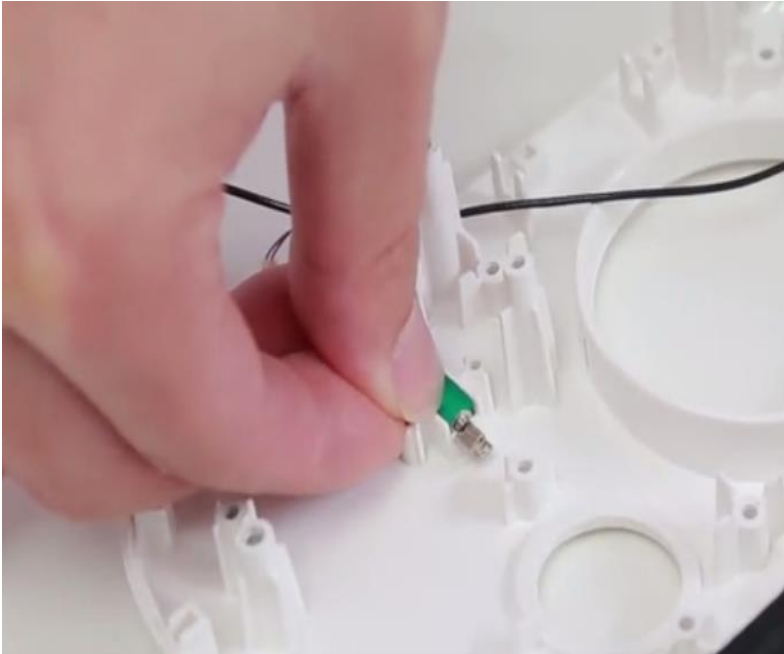
1. Use the Phillips screwdriver to remove the two screws securing the motor, then remove the motor module and mounting piece.

Disassembling the T16 Remote Controller

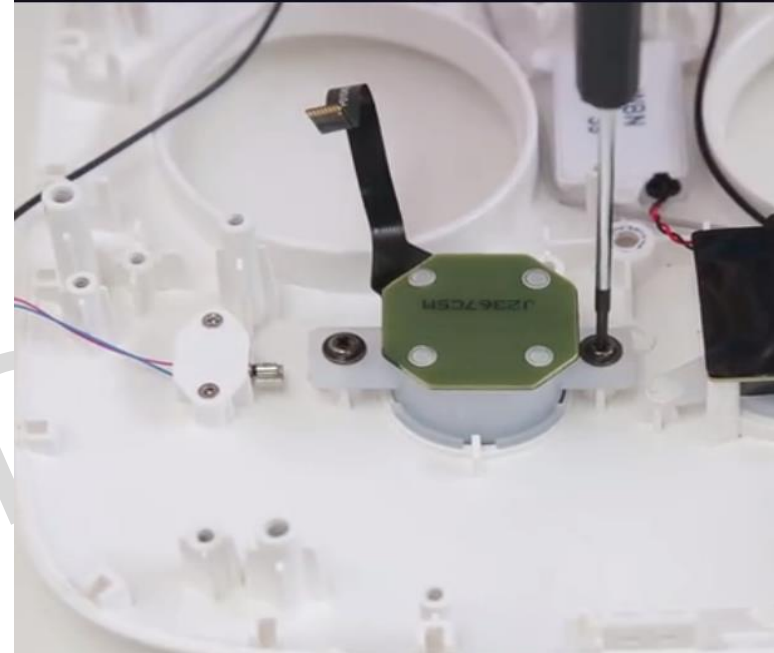
No.	Disassembly and Assembly	Picture	Notice
1	Use the Phillips screwdriver to remove the two screws securing the built-in battery cover, then remove the battery and cover.		
2	Use the Phillips screwdriver to remove the three screws securing the right dustproof module on the power board, then remove the module.		
3	Use the Phillips screwdriver to remove the four screws securing the built-in battery bracket on the power board, then remove the bracket.		
4	Disconnect the cable on the power board, use the Phillips screwdriver to remove the two screws securing the power board, and then remove the power board.		Check whether the power board is mounted in place.
5	Disconnect the cables on the battery port board and C3 button board. Use the Phillips screwdriver to remove the two screws securing the battery port board, and then remove the battery port board.		Check whether the power button board is mounted in place.
6	Use the Phillips screwdriver to remove the three screws securing the C3 button board, and remove the C3 button board.		Check whether the C3 button board is mounted in place.
7	Use the Phillips screwdriver to remove the two screws securing the USB port board, and remove the USB port board.		Check whether the USB port board is mounted in place.

No.	Notices	Picture
1	Ensure that the eight screws on the remote controller rear shell are removed.	
2	Press the middle of remote controller rear shell (there is a buckle inside) down vertically with the right hand thumb, and hold the handle with the other four fingers. Hold the screen bracket with the left hand and press down the right thumb with the left thumb. Then pull the handle to the right to separate the front and rear shell. Do not press down on the front shell.	
3	The buckle on the front shell is shown on the right figure. Do not press down on the front shell.	
4	The buckle on the rear shell is shown on the right figure. To separate the rear and front shells, press the middle of the remote controller rear shell.	
5	Press the middle of the remote controller rear shell as shown.	
6	Gently separate the front shell and rear shell. Otherwise, the flat cable may be damaged.	

Assembling the T16 Remote Controller



Mount the motor to the upper cover and mount the mounting piece, then tighten the two screws.

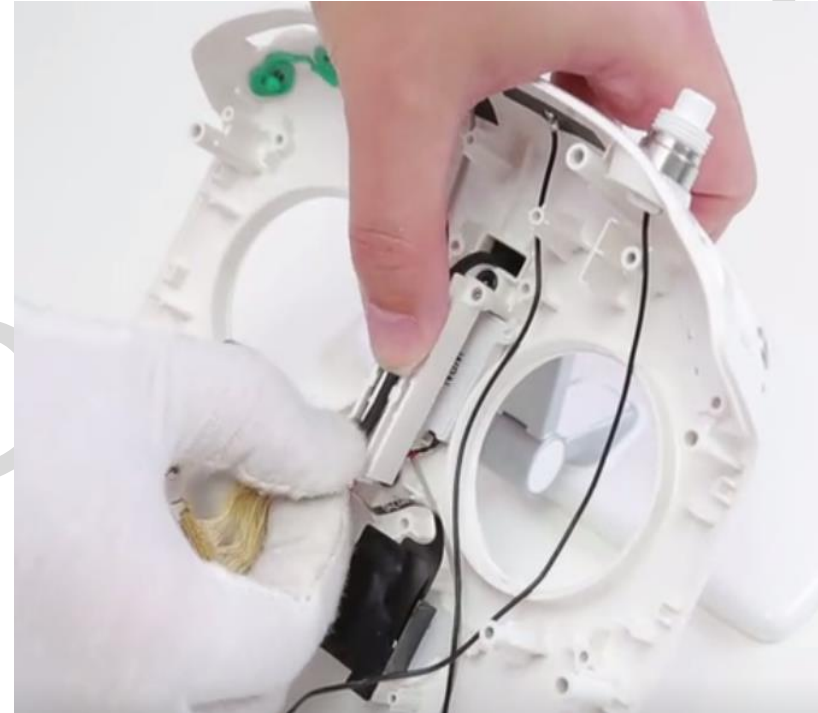


Mount the RTH button module and tighten the two screws.

Assembling the T16 Remote Controller



Pass the screen's flat cable through the upper shell module.
Mount the screen module and secure the screws.



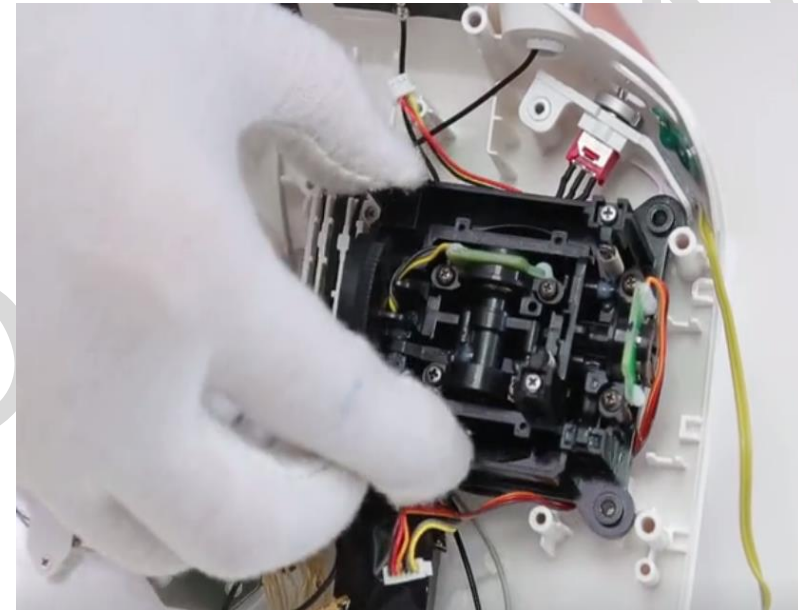
Fix the two antennas and screen's flat cable to the cable slot.
Ensure that the upper antenna port connects to the left antenna and lower port connects to the right antenna.
Differentiate between the upper and lower rubber stoppers.

Assembling the T16 Remote Controller



Mount the attitude lever module.

Check whether the A/B button module is mounted in place.

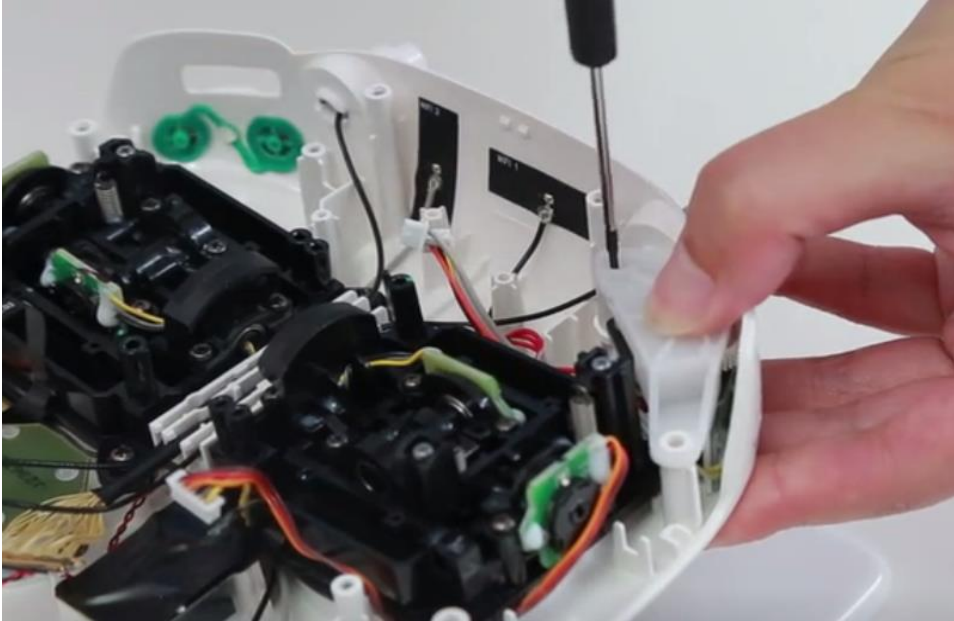


Mount the two control stick modules and secure each module with the four screws.

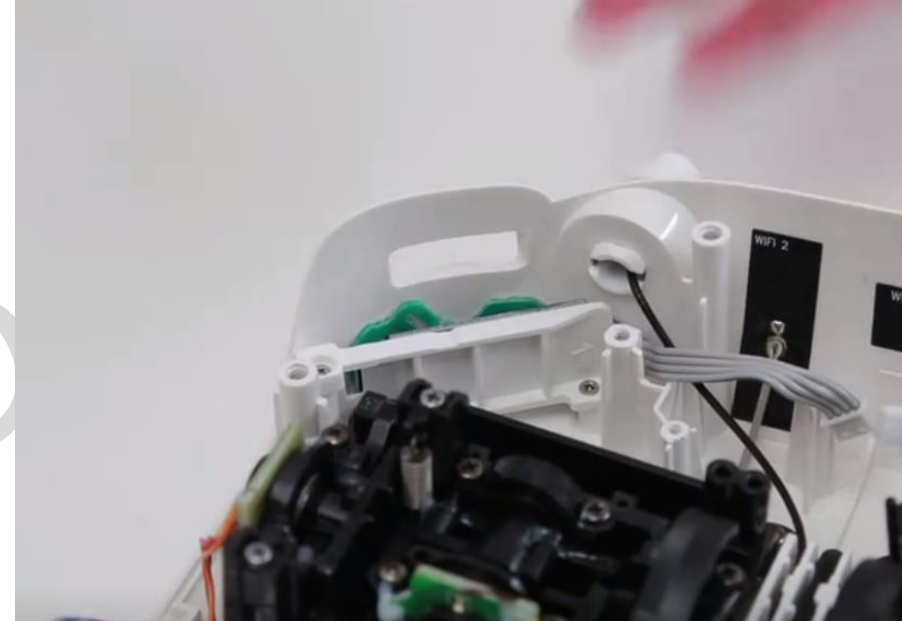
Note: 1. The small green board faces outwards.

2. Do not press down on the dustproof rubber ring and cable.

Assembling the T16 Remote Controller



Mount the spray rate dial module and tighten one screw.

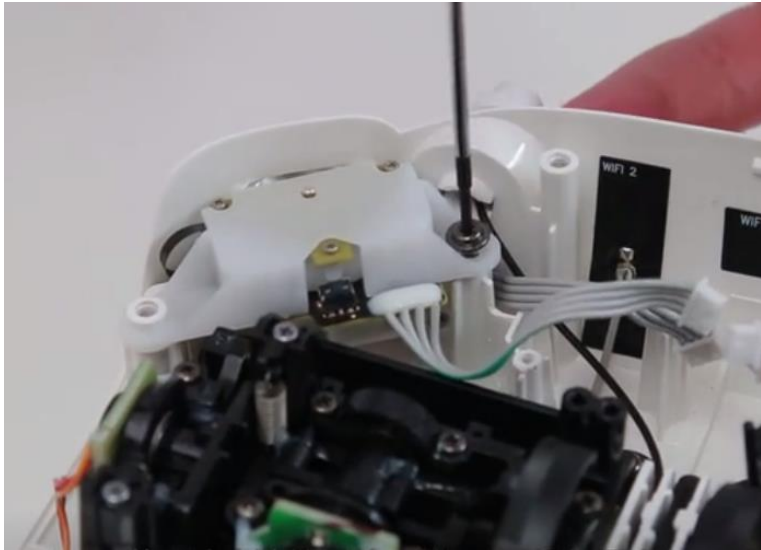


Mount the A/B button module and tighten the two screws.

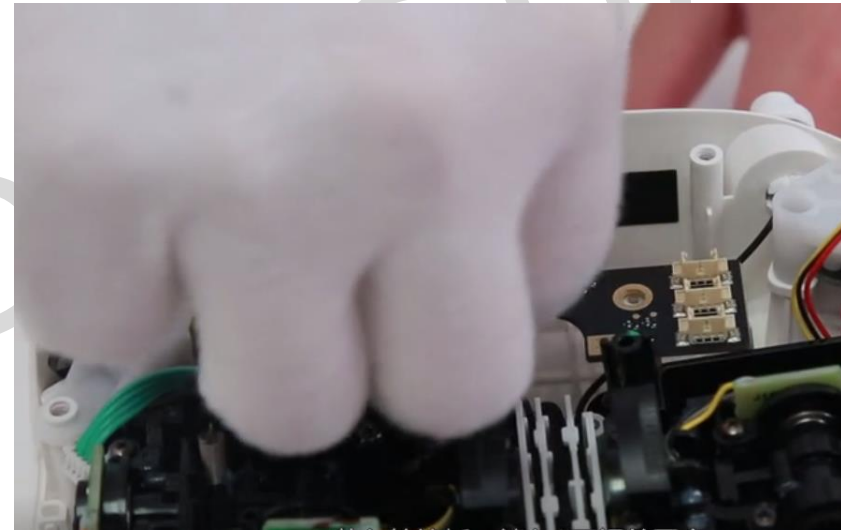
Pay attention to the installation direction, as the left and right control sticks are the same.

The sliding varistor is closer to the outside.

Assembling the T16 Remote Controller

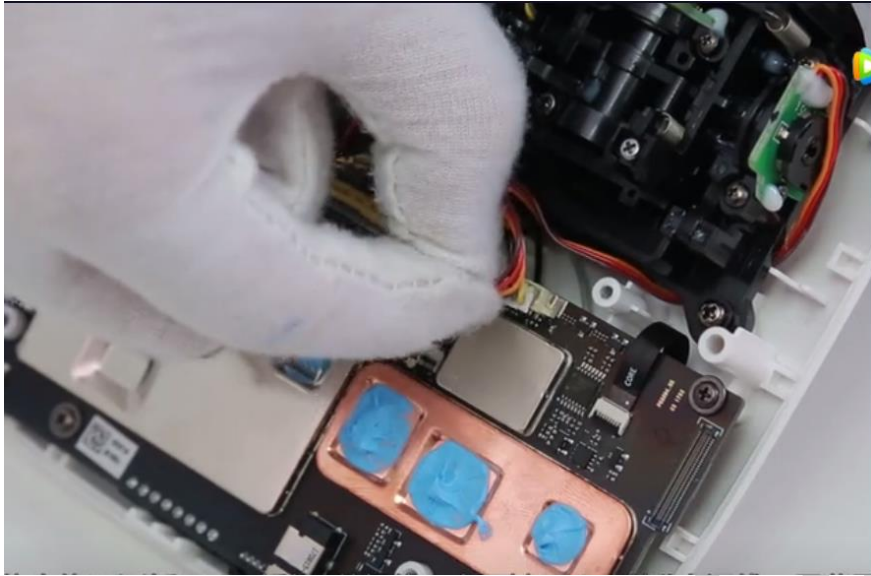


Mount the speed dial module and tighten the screw.
Check whether the speed dial is mounted in place.



Mount the adapter board and tighten the two screws.
Attach the cable and mount the button.

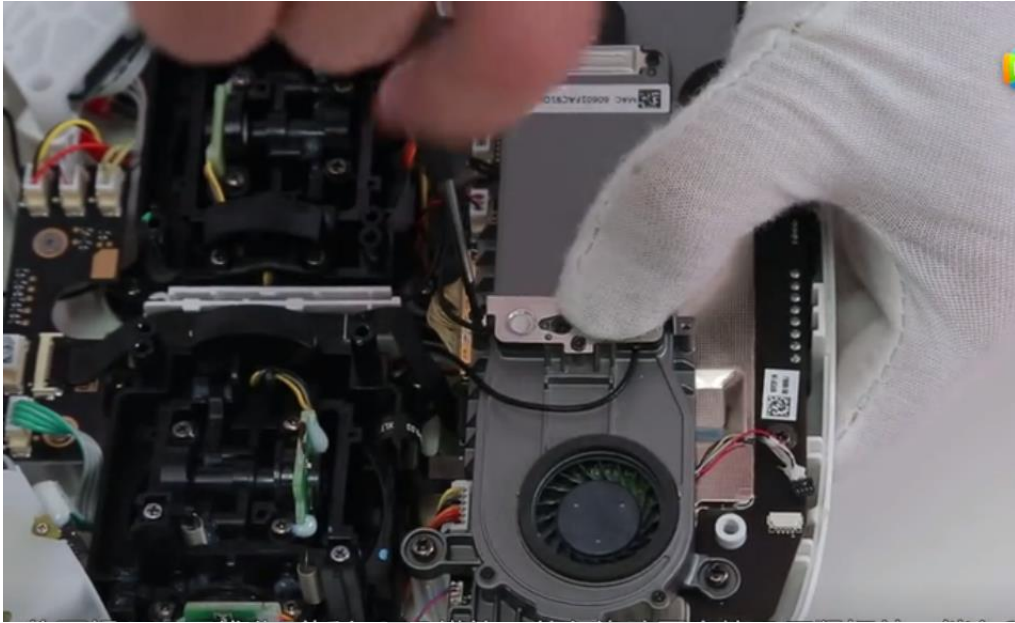
Assembling the T16 Remote Controller



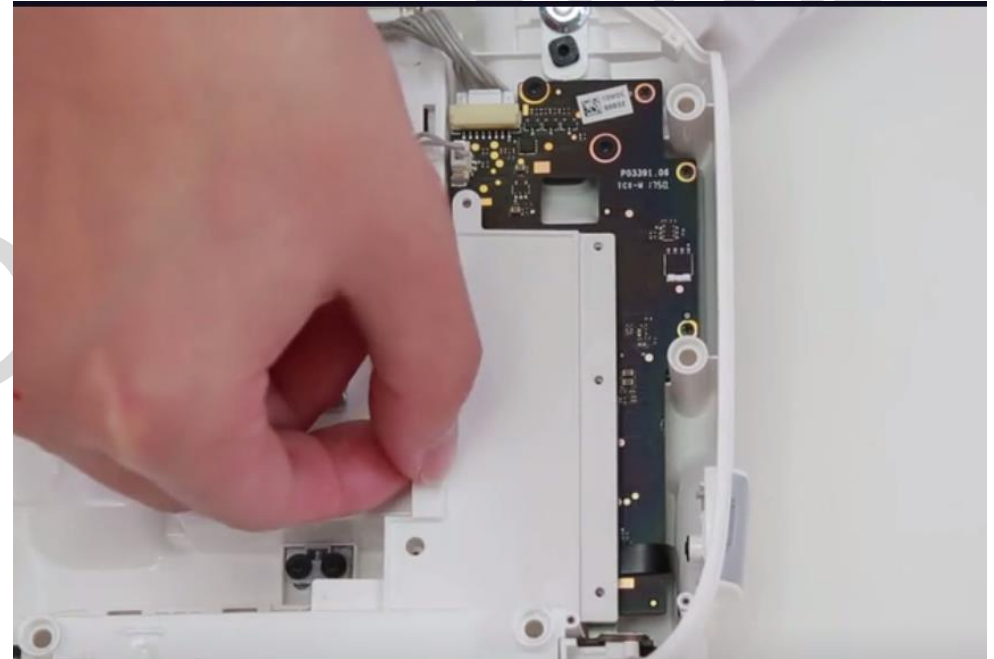
Attach the Wi-Fi board FPC, control stick 1 cable, power button FPC, speaker power cable, screen flat cable, cable which connects the main board and adapter board, RTH cable, control stick 2 cable, and motor cable.



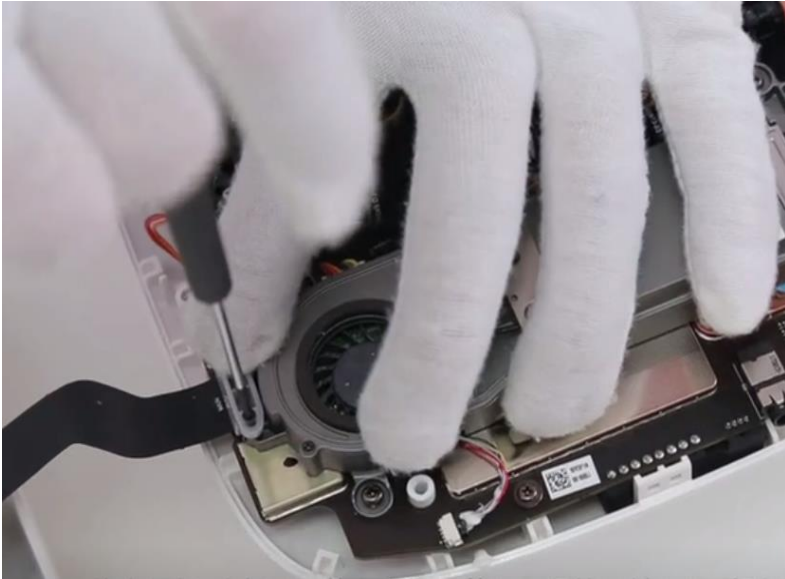
Mount the heat sink and SDR module, then tighten the four screws.



Attach the two SDR antennas to the SDR module. Mount the rubber and mounting piece, then tighten the three screws.



Mount the built-in battery compartment cover and tighten the four screws.



Attach the cable which connects the power board and main board to the main board. Mount the fixing piece and tighten the two screws. Ensure that the cable is attached in place.



Align and fasten the upper and lower shells and tighten the 8 screws. Attach the bottom silicone rubber, and check whether the buckles on the top and bottom of the shells are mounted in place and whether there is a gap between the front and rear shells.

Assembling the T16 Remote Controller



Align and fasten the upper and lower shells and tighten the 8 screws.
Attach the rubber pad.



Attach the two antennas and rotate the metal ring to tighten it. Insert the antennas, and then tighten the metal ring.

T16 Remote Controller Troubleshooting



Module	Issue	Cause Analysis	Solution
Remote Controller	The remote controller cannot be turned on.	Check the connection between the battery port board, power board, remote controller main board, and power button, and check whether the cables are connected well.	Replace the defective component.
	DJI Assistant 2 cannot be connected.	1. The version of DJI Assistant 2 is not compatible with the product.	Download the corresponding version of DJI Assistant 2 on DJI's official website and update the firmware version to the latest.
		2. The driver installation failed.	1. Reinstall the DJI Assistant 2 software. 2. Use a different the computer.
		3. The main board is damaged.	Replace the main board.
	Linking failure	1. Check whether the remote controller firmware matches the drone's firmware.	1. Update the remote controller firmware to the latest. 2. Update the aircraft firmware to the latest.
		2. The SDR board is damaged.	Replace the RF board.
		3. The main board is damaged.	Replace the main board.
	The remote controller signal is lost within the specified flight range.	1. The antenna is damaged.	Replace the antenna.
		2. The SDR board is damaged.	Replace the RF board.
		3. The main board is damaged.	Replace the main board.
	The screen is malfunctioning.	The screen or coaxial cable is broken.	Replace the screen.
	Satellite search is abnormal.	The remote controller screen is damaged.	Replace the screen.
	Wi-Fi cannot be connected.	1. Check whether the Wi-Fi antenna is damaged or not connected well.	Replace or reattach the Wi-Fi antenna.
		2. The Wi-Fi board is damaged.	Replace the Wi-Fi board.
		3. The main board is damaged.	Replace the main board.
	The button is malfunctioning.	The button is damaged.	Replace the button.
		The adapter board is damaged.	Replace the adapter board.
		The main board is damaged.	Replace the main board.

T16 Remote Controller Troubleshooting

1. How can the remote controller be disassembled quickly and safely?

- 1) Remove the eight screws on the remote controller.
- 2) Knock on the area shown in Figure 2 with the tool shown in Figure 1 to open the remote controller.



Figure 1



Figure 2

2. The remote controller controllable range is short.

Check whether the remote controller antenna is broken. If yes, replace it properly. (Pay special attention to the antennas in the three positions marked with the red boxes as shown in Figures 3 and 4.)



Figure 3

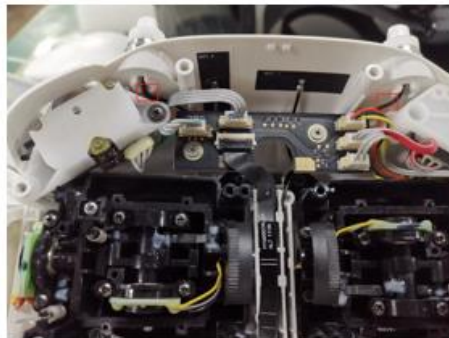


Figure 4

3. The remote controller buttons don't make any noise.

1. Check whether the buttons on the remote controller are functioning.
2. Replace the remote controller's main board.



Figure 5

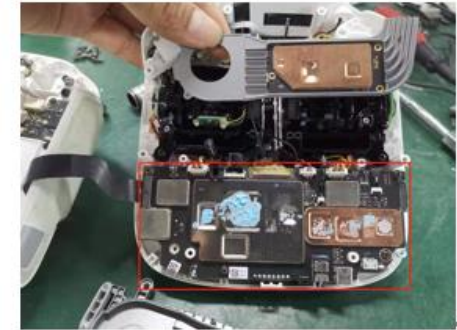


Figure 6

4. The remote controller control sticks cannot be calibrated (when calibrating the control sticks, the percentage does not change or cannot reach 100%).

1. Restore the remote controller to the factory settings.
2. Replace the remote controller control sticks.



Figure 7

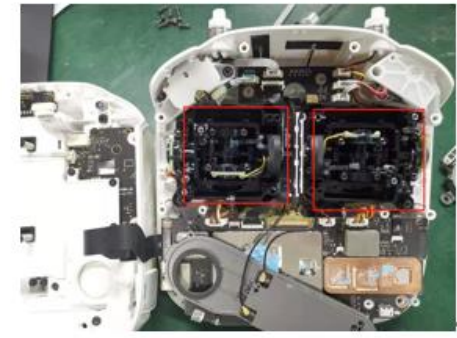


Figure 8

5. The control stick cannot center (The control stick cannot center after calibration, and the app prompts a warning stating that the control stick does not center).

1. Restore the remote controller to the factory settings.
2. Replace the remote controller main board.



Figure 9



Figure 10

6. The app prompts that the remote controller's temperature is too high.

1. Check whether the air inlet and outlet are blocked (the air inlet on the right and the heat dissipation on the left);
2. Check whether the cooling fan of the remote controller works normally. If not, replace it. (Replace the fan; pay attention to clean up the remaining glue, otherwise the fan will be uneven.)



Figure 11



Figure 12

7. The remote controller cannot be linked with the aircraft.

Replace the SDR board (if the remote controller is malfunctioning).



Figure 13

8. The remote controller does not make alarms, or the warning tone sounds abnormal.

Replace the remote controller's speaker (the speaker is fixed in place by the metal rod on the front shell).



Figure 14



Figure 15

9. The remote controller's built-in battery cannot be charged, or the remote controller's data protection function does not work.

1. Install the built-in battery to the remote controller again.
2. Replace the built-in battery.



Figure 16



Figure 17

10. The spray dial does not work when performing the second step of the remote controller calibration.

1. Replace the camera dial module on the remote controller.
2. Replace the adaptive board.
3. Replace the flat cable.



Figure 18



Figure 19

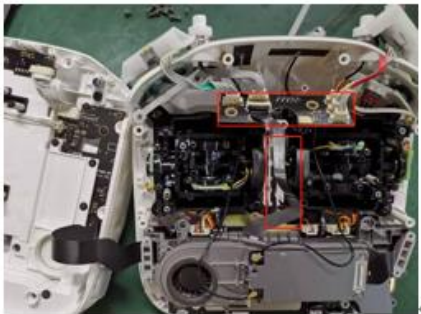


Figure 20

**THANK
YOU**